

Bhavana Devulapally

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EDUCATION

University of Pittsburgh, United States

Apr 2025

Master of Science in Information Science

GPA: 3.8/4.0

Osmania University, India

Jul 2023

Bachelor of Engineering in Computer Science

GPA: 8.7/10.0

WORK EXPERIENCE

Software Engineer | NCR Corporation, Hyderabad, India

Feb '23 - July '23

Cash Services- Self-Checkout Application (Python, ReactJS, Docker, Kubernetes)

- Developed a user-friendly self-checkout application, driving improvements in the cash services sector.
- Containerized the application by creating **Docker images** and created a deployment pipeline to deploy the application on self-hosted machines. Made the deployment easy to manage by integrating the pipeline with Ansible playbooks.
- Reduced the deployment processing time by 27% by using **Kubernetes** for efficient cluster management.

Shared Template - Task Standardization and Performance Enhancement (Git, Jenkins)

- Improved the backend performance by **32%** by using a shared template to streamline task execution across multiple jobs.
- Enhanced the latency of the workload by 75%, which helped the partnering clients Walmart, Toshiba and CaseysUSA.
- Added testcases and increased the test coverage to 90%. This **reduced the bugs in production by 50%**, which reduced the burden on the on-call rotations. Prepared detailed technical documentation supporting deployment and system processes.

Software Developer | ADP Private Limited, Hyderabad, India

Jun '22 – Dec '22

Real-time Deployment Status Tracker (Angular, Django)

- Developed a web application to **monitor team deployment status**, displaying business critical metric details such as health check status, configuration versions, image versions, and logs. Optimized information flow reducing manual effort for SRE team members by automating checks.
- Integrated APIs** for seamless backend data retrieval, enhancing real-time updates on the frontend.
- Enhanced operational efficiency by 81% by providing real-time deployment insights, boosting team productivity.

PROJECT EXPERIENCE

Scalable Log & Text Data Processor (Python, Hadoop, YARN, HDFS, Docker, Ubuntu) - [Link](#)

- Deployed a **3-node HDFS cluster** using Docker, cutting data processing time by 90% and ensuring fault-tolerant log analysis.
- Designed and fine-tuned **Python-based MapReduce jobs** to analyze 5GB of server logs, generating a flexible n-gram frequency model and extracting 10 key insights, including IP requests, HTTP response codes, and frequently accessed paths.

YouTube Content Safety Monitor (ML, NLTK, Python, BERT, YouTube Data API v3) - [Link](#)

- Developed a **BERT-powered video classification model** with built-in profanity detection and topic analysis using NLTK, achieving **83% accuracy for enhanced parental monitoring** of YouTube content.
- Built a scalable data ingestion pipeline by capturing network traffic through a **Proxy with SSL interception**, indexing over 10,000 logs into **ElasticSearch** for efficient content tracking.

Predictive Paint Color Analysis (PPG Industries) - [Link](#)

- Developed and optimized regression and classification models to accurately forecast paint characteristics and consumer preferences using RGB and HSL color space data.
- Employed advanced techniques like **hyperparameter tuning**, **cross-validation**, and feature selection within the caret framework in R for robust model performance. Got an accuracy of 92% by refining the model using the **LOGIT-transformed response**.

Center for Comparative Archaeology Website (Flask, MySQL, HTML, JavaScript) - [Link](#)

- Built a **microservices-based** archaeology database system, boosting collaborative data access by 55% and streamlining contributions.
- Designed scalable and efficient schemas for SQL (MySQL) and NoSQL (MongoDB) databases, supporting CRUD operations for researchers.
- Enabled accurate data tracking with timestamped entries and advanced filtering, improving retrieval efficiency by 45%.

TECHNICAL SKILLS

Programming & Web Languages

Python, JAVA, R, React, Node.js, Vue.js, Express.js, Angular, HTML, PHP, AJAX

Database

MySQL, MongoDB, SQL, Cassandra

Libraries & Frameworks

Django, Flask, Pandas, NumPy, matplotlib, Git, Rest API's, XML

DevOps & CI/CD

Kubernetes, Docker, Jenkins, Ansible

Certifications

Top 50 Worldwide in the CodeChef LunchBox Hackathon, Introduction to ML – NPTEL IIT Madras, [“The Joy of Computing Using Python” conducted by NPTEL \(2019\)](#)